Model-Implied Instrumental-Variable and James–Stein Estimation of Structural Equation Models: A Case Study With lavaan

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Abstract

This article presents model-implied instrumental-variable (MIIV) estimation and scaling (JS) and aggregated (JSA) James–Stein estimation within a common notation, based on their implementation in the R package lavaan. We describe the estimation of factor loadings and latent regression slopes equation by equation, followed by the estimation of variance, covariance, and intercept parameters. We also provide analytical standard errors for JS and JSA using the delta method, accounting for uncertainty across all estimation stages. A simulation study compares the standard error accuracy of these approaches with maximum likelihood (ML) under correct model specification. The standard errors of MIIV, JS, and JSA perform comparably to those of ML under the conditions examined.

Keywords: structural equation modeling, model-implied instrumental variables, James–Stein estimation, delta method, simulation

Structural equation models (SEMs) with latent variables are usually estimated by maximum likelihood (ML), which estimates all parameters jointly. Although ML has desirable asymptotic properties when its assumptions hold, difficulties can arise when it is applied in practice. For instance, a misspecification in one equation can bias estimates in other equations (Bollen et al., 2007; Bollen, 2019; Dhaene & Rosseel, 2023; Kaplan, 1988; see also Jin et al., 2021, for diagonally weighted least squares), and the iterative optimization can fail to converge or end in inadmissible solutions, especially in small samples (Bollen, 2019; Dhaene & Rosseel, 2023). Equation-by-equation estimators offer an alternative (e.g., Bollen, 1996; Burghgraeve et al., 2021). In the approaches considered, factor loadings and latent regression slopes are estimated

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one equation at a time and in closed form, which may limit the propagation of misspecification across equations (Bollen et al., 2018; Burghgraeve et al., 2021) and avoids iterative optimization for this part of the estimation. Two such estimators are considered. Model-implied instrumental-variable (MIIV) estimation replaces each latent variable by a scaling indicator and uses instruments implied by the model to correct for the resulting measurement error (Bollen, 1996, 2019; Bollen & Bauer, 2004). James–Stein estimation, proposed for SEMs by Burghgraeve et al. (2021), instead replaces each latent predictor by an estimate of its conditional expectation given observed proxies and then applies least squares. This conditional expectation is estimated using James–Stein type shrinkage (Efron & Morris, 1973, 1975; James & Stein, 1961). The approach follows the principle of regression calibration in errors-in-variables regression (Carroll et al., 2006; Fuller, 1987; Whittemore, 1989). Its scaling version (JS) uses the scaling indicators as proxies, and its aggregated version (JSA) uses reliability weighted sums of indicators. JSA was found to outperform ML and MIIV in terms of mean squared error under the small sample conditions examined (Burghgraeve et al., 2021). Both versions were further evaluated for confirmatory factor models by Dhaene and Rosseel (2024).

Beyond point estimation, the usefulness of these approaches for statistical inference depends on the accuracy of their standard errors. For MIIV, conventional standard errors for factor loadings and latent regression slopes follow from two-stage least squares (Bollen, 1996). Standard errors for these same estimates can also be obtained by the delta method, using derivatives with respect to the sample moments, for which Fisher and Bollen (2020) provided explicit analytical expressions. These derivatives also propagate uncertainty in the estimated loadings and slopes into the standard errors of subsequently estimated variance and covariance parameters. Standard errors for these latter parameters were derived under unweighted least squares by Bollen and Maydeu-Olivares (2007) and extended to general weight matrices by Jin et al. (2021). More recently, Maydeu-Olivares et al. (in press) compared subsequent variance and covariance estimation procedures and the accuracy of their standard errors. For James–Stein estimation, Burghgraeve et al. (2021) derived a sampling variance expression for latent regression estimates that accounts for uncertainty in the estimated reliability adjustment, providing a basis for analytical standard errors for those estimates. However, they did not develop a complete analytical standard error calculation for all model parameters and used bootstrap standard errors for both JS and JSA in their empirical evaluations. The R package lavaan (Rosseel, 2012) now provides analytical standard errors for both estimators by applying the delta method to the complete estimation procedure. In addition, its JS and JSA implementations, initially restricted to confirmatory factor models, have been extended to more general models, and the package also includes MIIV estimation.

Present Study and Contributions

This article makes three minor contributions. First, it describes and conceptually compares MIIV and James–Stein estimation within a common notation, following their implementation in lavaan. Second, it develops and illustrates a complete delta method calculation of standard errors for JS and JSA, accounting for uncertainty across all estimation stages. Third, it evaluates their standard error accuracy in a simulation study comparing JS and JSA with MIIV and ML.

The remainder of the article introduces the model and notation, describes the estimation of factor loadings, latent regression slopes, and the remaining parameters, presents the analytical standard errors, and reports the simulation study.

Model, Notation, and Parameterization

Following the all-y LISREL notation (Jöreskog & Sörbom, 1996, submodel 3B, Chapter 6; Rosseel & Vidal, 2025), let $\mathbf{y} \in \mathbb{R}^p$ contain the continuous observed variables and $\boldsymbol{\eta} \in \mathbb{R}^q$ the latent variables. The measurement and latent variable models are

$$\mathbf{y} = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\eta} = \boldsymbol{\alpha} + \mathbf{B}\boldsymbol{\eta} + \boldsymbol{\zeta},$$

where $\boldsymbol{\nu} \in \mathbb{R}^p$ contains the measurement intercepts, $\boldsymbol{\Lambda} \in \mathbb{R}^{p\times q}$ contains the factor loadings, $\boldsymbol{\alpha} \in \mathbb{R}^q$ contains the structural intercepts, and $\mathbf{B} \in \mathbb{R}^{q\times q}$ contains the latent regression slopes, that is, the regression slopes among the latent variables. The measurement errors $\boldsymbol{\epsilon} \in \mathbb{R}^p$ and structural disturbances $\boldsymbol{\zeta} \in \mathbb{R}^q$ satisfy $E(\boldsymbol{\epsilon}) = \mathbf{0}$ and $E(\boldsymbol{\zeta}) = \mathbf{0}$. Let $\mathbb{S}_+^k$ and $\mathbb{S}_{++}^k$ denote the sets of symmetric positive semidefinite and positive definite $k \times k$ matrices, respectively. Assume

$$\mathrm{Var}(\boldsymbol{\epsilon}) = \boldsymbol{\Theta} \in \mathbb{S}_+^p, \qquad \mathrm{Var}(\boldsymbol{\zeta}) = \boldsymbol{\Psi} \in \mathbb{S}_+^q, \qquad \mathrm{Cov}(\boldsymbol{\epsilon}, \boldsymbol{\zeta}) = \mathbf{0}_{p\times q}.$$

Provided that $\mathbf{I}_q - \mathbf{B}$ is nonsingular, the latent and observed mean vectors are

$$\boldsymbol{\mu}_\eta = (\mathbf{I}_q - \mathbf{B})^{-1}\boldsymbol{\alpha} \in \mathbb{R}^q, \qquad \boldsymbol{\mu}_y = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\mu}_\eta \in \mathbb{R}^p,$$

where $\mathbf{I}_q$ is the $q \times q$ identity matrix. The implied observed covariance matrix is

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}(\mathbf{I}_q - \mathbf{B})^{-1}\boldsymbol{\Psi}(\mathbf{I}_q - \mathbf{B})^{-\top}\boldsymbol{\Lambda}^\top + \boldsymbol{\Theta}.$$

Let $\boldsymbol{\theta} \in \mathbb{R}^d$ collect the free parameters. Assume that $\boldsymbol{\Sigma}(\boldsymbol{\theta}) \in \mathbb{S}_{++}^p$. Denote the sample mean vector and covariance matrix by $\bar{\mathbf{y}} \in \mathbb{R}^p$ and $\mathbf{S} \in \mathbb{S}_+^p$, respectively, computed from a single group of independent and identically distributed observations $\mathbf{y}_i$, $i = 1, \ldots, n$, of $\mathbf{y}$, which is the setting of all estimation expressions below.[1] Define

$$\mathbf{s} = \begin{pmatrix} \bar{\mathbf{y}} \\ \mathrm{vech}(\mathbf{S}) \end{pmatrix} \in \mathbb{R}^h, \qquad \boldsymbol{\sigma}(\boldsymbol{\theta}) = \begin{pmatrix} \boldsymbol{\mu}_y(\boldsymbol{\theta}) \\ \mathrm{vech}\{\boldsymbol{\Sigma}(\boldsymbol{\theta})\} \end{pmatrix} \in \mathbb{R}^h, \qquad h = p + \frac{p(p+1)}{2},$$

where vech stacks the lower triangular entries, including the diagonal.

Estimation of Factor Loadings and Latent Regression Slopes

JS, JSA, and MIIV estimate factor loadings and latent regression slopes equation by equation, under a scaling indicator parameterization (Bollen, 1996, 2019; Burghgraeve et al., 2021), in which one loading per latent variable, that of its scaling indicator, is fixed to one and the corresponding intercept is fixed to zero (Bollen, 1989). These estimates are obtained from centered observations, with intercepts estimated subsequently. The following example illustrates how JS and MIIV address measurement error in an observed predictor.

Example 1. Consider four indicators of η_1, with y_1 as its scaling indicator:

$$y_1 = \eta_1 + \epsilon_1, \qquad y_r = \nu_r + \lambda_r\eta_1 + \epsilon_r, \quad r = 2, 3, 4.$$

Assume

$$E(\epsilon_r) = 0, \qquad \mathrm{Cov}(\eta_1, \epsilon_r) = 0, \qquad \mathrm{Cov}(\epsilon_r, \epsilon_s) = 0 \quad (r \neq s).$$

The objective is to estimate λ_2. Writing $\psi = \mathrm{Var}(\eta_1)$ and $\omega_1 = \mathrm{Var}(\epsilon_1)$, the population slope from regressing y_2 directly on y_1 is

$$\frac{\mathrm{Cov}(y_1, y_2)}{\mathrm{Var}(y_1)} = \lambda_2\frac{\psi}{\psi + \omega_1}.$$

Thus, measurement error in y_1 attenuates the slope.

JS estimates the latent predictor from its scaling indicator. Under joint normality,

$$E(\eta_1 \mid y_1) = \mu_1 + R(y_1 - \mu_1), \qquad R = \frac{\psi}{\psi + \omega_1},$$

[1] Following lavaan, $\mathbf{S}$ uses the divisor n for JS and JSA and $n - 1$ for MIIV estimation.

where $\mu_1 = E(\eta_1) = E(y_1)$. The other indicators provide information about ω_1, because

$$\mathrm{Cov}(y_1, y_3) = \lambda_3\psi, \qquad \mathrm{Cov}(y_1, y_4) = \lambda_4\psi, \qquad \mathrm{Cov}(y_3, y_4) = \lambda_3\lambda_4\psi.$$

Consequently,

$$\psi = \frac{\mathrm{Cov}(y_1, y_3)\,\mathrm{Cov}(y_1, y_4)}{\mathrm{Cov}(y_3, y_4)}, \qquad \widehat{\omega}_1 = s_{11} - \frac{s_{13}s_{14}}{s_{34}},$$

where s_{rs} denotes a sample covariance. This illustrates the preliminary error variance calculation using one pair of additional indicators.

With the small sample correction used in JS, the estimated reliability adjustment and fitted latent predictor are

$$\widehat{R} = \frac{s_{11} - c_n\widehat{\omega}_1}{s_{11}}, \qquad c_n = \frac{n-3}{n-1}, \qquad \widetilde{\eta}_1 = \overline{y}_1 + \widehat{R}(y_1 - \overline{y}_1).$$

Regressing y_2 on $\widetilde{\eta}_1$, with an intercept, gives

$$\widehat{\lambda}_{2,\mathrm{JS}} = \frac{\widehat{R}\,s_{12}}{\widehat{R}^2 s_{11}} = \frac{s_{12}}{s_{11} - c_n\widehat{\omega}_1}.$$

MIIV instead substitutes $\eta_1 = y_1 - \epsilon_1$ into the equation for y_2:

$$y_2 = \nu_2 + \lambda_2 y_1 + u, \qquad u = \epsilon_2 - \lambda_2\epsilon_1.$$

The explanatory variable y_1 correlates with the composite error,

$$\mathrm{Cov}(y_1, u) = -\lambda_2\omega_1,$$

whereas y_3 satisfies $\mathrm{Cov}(y_3, u) = 0$. It can therefore serve as an instrument when $\mathrm{Cov}(y_3, y_1) \neq 0$, giving

$$\mathrm{Cov}(y_3, y_2) = \lambda_2\,\mathrm{Cov}(y_3, y_1), \qquad \widehat{\lambda}_{2,\mathrm{MIIV}} = \frac{s_{32}}{s_{31}}.$$

With several instruments, such as y_3 and y_4, two-stage least squares first regresses y_1 on the instruments and then regresses y_2 on the fitted values, including intercepts in both regressions.

For either estimator,

$$\widehat{\nu}_2 = \overline{y}_2 - \widehat{\lambda}_2\overline{y}_1.$$

JS therefore adjusts the predictor using an estimated measurement error variance, whereas MIIV uses instruments to estimate the loading despite the correlation between the explanatory variable and the composite error. JSA follows the JS calculation using a normalized aggregate of eligible indicators in place of y_1.

James–Stein Estimation

Consider one measurement or latent variable equation. Let $\mathbf{f} \in \mathbb{R}^k$ collect its latent predictors and $\mathbf{a} \in \mathbb{R}^k$ the corresponding factor loadings or latent regression slopes. Let $\mathbf{x} \in \mathbb{R}^k$ contain their scaling indicators in the same order, and let $\mathbf{e} \in \mathbb{R}^k$ contain the corresponding measurement errors. The scaling indicator rows of the measurement model give

$$\mathbf{x} = \mathbf{f} + \mathbf{e},$$

which generalizes $y_1 = \eta_1 + \epsilon_1$ in the example. Write

$$\boldsymbol{\mu} = E(\mathbf{f}) = E(\mathbf{x}) \in \mathbb{R}^k, \qquad \boldsymbol{\Sigma}_x = \mathrm{Var}(\mathbf{x}) \in \mathbb{S}^k_{++}, \qquad \boldsymbol{\Omega} = \mathrm{Var}(\mathbf{e}) \in \mathbb{S}^k_{+}.$$

The observed dependent variable is the indicator being modeled in a measurement equation, such as y_2 in the example. For a latent variable equation, the dependent latent variable is replaced by its scaling indicator. Its error then contains both the structural disturbance and that scaling indicator's measurement error. Denote the resulting equation error by ε.

The JS procedures considered use proxies that load only on their respective latent predictors and have mutually uncorrelated measurement errors, so $\mathbf{\Omega}$ is diagonal. The required orthogonality conditions are

$$\mathrm{Cov}(\mathbf{f}, \mathbf{e}) = \mathbf{0}, \qquad \mathrm{Cov}(\mathbf{f}, \varepsilon) = \mathbf{0}, \qquad \mathrm{Cov}(\mathbf{e}, \varepsilon) = \mathbf{0}.$$

Under joint normality,

$$E(\mathbf{f} \mid \mathbf{x}) = \boldsymbol{\mu} + \mathbf{R}(\mathbf{x} - \boldsymbol{\mu}), \qquad \mathbf{R} = (\mathbf{\Sigma}_x - \mathbf{\Omega})\mathbf{\Sigma}_x^{-1} \in \mathbb{R}^{k \times k}.$$

Without normality, this expression is the best linear predictor of $\mathbf{f}$ from $\mathbf{x}$. JS estimates this predictor and then applies least squares (Burghgraeve et al., 2021).

Let $\mathbf{S}_x \in \mathbb{S}^k_{++}$ contain the sample covariances of the proxies and $\widehat{\mathbf{\Omega}} \in \mathbb{R}^{k \times k}$ be the diagonal matrix of preliminary measurement error variance estimates. The estimated reliability adjustment and fitted latent predictors are

$$\widehat{\mathbf{R}} = (\mathbf{S}_x - c_n\widehat{\mathbf{\Omega}})\mathbf{S}_x^{-1}, \qquad c_n = \frac{n-3}{n-1},$$

$$\widetilde{\mathbf{f}} = \overline{\mathbf{x}} + \widehat{\mathbf{R}}(\mathbf{x} - \overline{\mathbf{x}}).$$

Let $\mathbf{X} \in \mathbb{R}^{n \times k}$ contain the centered proxy observations and $\mathbf{d} \in \mathbb{R}^n$ the centered observations of the dependent variable. The centered fitted predictor matrix is

$$\widetilde{\mathbf{X}} = \mathbf{X}\widehat{\mathbf{R}}^\top.$$

The estimator is

$$\widehat{\mathbf{a}} = (\widetilde{\mathbf{X}}^\top\widetilde{\mathbf{X}})^{-1}\widetilde{\mathbf{X}}^\top\mathbf{d} = (\mathbf{S}_x - c_n\widehat{\mathbf{\Omega}})^{-1}\mathbf{s}_{xd},$$

where $\mathbf{s}_{xd} \in \mathbb{R}^k$ contains the sample covariances between the proxies and the dependent variable. The second expression shows that the fitted predictor regression is equivalent to subtracting the estimated measurement error contribution from the proxy covariance matrix. These expressions assume that $\widetilde{\mathbf{X}}$ has full column rank and that the loadings and slopes are not subject to equality constraints.

Scaling James–Stein Estimation

For scaling JS, $\mathbf{x}$ contains the scaling indicators of the latent predictors. Preliminary measurement error variances are obtained from the covariance relationships illustrated in the example. For an indicator y_r, two other eligible indicators y_s and y_t of the same latent variable give

$$\widehat{\omega}_r = s_{rr} - \frac{s_{rs}s_{rt}}{s_{st}}.$$

When several eligible pairs are available, the corresponding estimates are averaged. The preliminary estimates for the scaling indicators form $\widehat{\mathbf{\Omega}}$, yielding

$$\widehat{\mathbf{a}}_{\mathrm{JS}} = (\mathbf{S}_x - c_n\widehat{\mathbf{\Omega}})^{-1}\mathbf{s}_{xd}.$$

These preliminary error variance estimates are used for prediction and need not equal the final model estimates.

Aggregated James–Stein Estimation

JSA replaces each scaling indicator by a reliability weighted aggregate of indicators of the same latent predictor (Burghgraeve et al., 2021). The indicators must satisfy the proxy conditions above, which exclude the dependent indicator when estimating a measurement equation.

For each latent predictor f in $\mathbf{f}$, let $\mathbf{y}_f \in \mathbb{R}^r$ collect its eligible indicators, with corresponding intercepts $\boldsymbol{\nu}_f \in \mathbb{R}^r$ and loadings $\boldsymbol{\lambda}_f \in \mathbb{R}^r$. For weights $\mathbf{w} \in \mathbb{R}^r$, normalizing the aggregate to the latent variable's scale gives

$$g = \frac{\mathbf{w}^\top(\mathbf{y}_f - \boldsymbol{\nu}_f)}{\mathbf{w}^\top\boldsymbol{\lambda}_f} = f + e_g,$$

where e_g is the aggregate's measurement error. Let $\mathbf{Q}_f \in \mathbb{S}_+^r$ be the diagonal covariance matrix of the selected indicators' measurement errors. Then

$$\omega_g = \mathrm{Var}(e_g) = \frac{\mathbf{w}^\top\mathbf{Q}_f\mathbf{w}}{(\mathbf{w}^\top\boldsymbol{\lambda}_f)^2}.$$

Let $\widehat{\mathbf{Q}}_f$ contain preliminary measurement error variance estimates and $\mathbf{S}_f \in \mathbb{S}_{++}^r$ the sample covariance matrix of these indicators. The weights maximize estimated reliability:

$$\widehat{\mathbf{w}} \in \arg\max_{\mathbf{w}\neq\mathbf{0}} \left(1 - \frac{\mathbf{w}^\top\widehat{\mathbf{Q}}_f\mathbf{w}}{\mathbf{w}^\top\mathbf{S}_f\mathbf{w}}\right).$$

Using preliminary scaling JS loadings $\widehat{\boldsymbol{\lambda}}_f$, the centered aggregate and its estimated error variance are

$$\widehat{g} = \frac{\widehat{\mathbf{w}}^\top(\mathbf{y}_f - \overline{\mathbf{y}}_f)}{\widehat{\mathbf{w}}^\top\widehat{\boldsymbol{\lambda}}_f}, \qquad \widehat{\omega}_g = \frac{\widehat{\mathbf{w}}^\top\widehat{\mathbf{Q}}_f\widehat{\mathbf{w}}}{(\widehat{\mathbf{w}}^\top\widehat{\boldsymbol{\lambda}}_f)^2},$$

with nonzero normalization denominator.

Constructing one aggregate for each latent predictor gives the columns of $\mathbf{X}$, and their estimated error variances form the diagonal of $\widehat{\boldsymbol{\Omega}}$, in the same order. The aggregate covariance matrix and its covariances with the dependent variable replace $\mathbf{S}_x$ and $\mathbf{s}_{xd}$ in the preceding expressions, giving

$$\widehat{\mathbf{a}}_{\mathrm{JSA}} = (\mathbf{S}_x - c_n\widehat{\boldsymbol{\Omega}})^{-1}\mathbf{s}_{xd}.$$

Thus, JS and JSA share the estimation formula and differ in the construction of the proxies and their measurement error variances.

Model-Implied Instrumental-Variable Estimation

MIIV uses the scaling indicator equations to replace latent variables with observed variables (Bollen, 1996). For the same equation considered above, substituting

$$\mathbf{f} = \mathbf{x} - \mathbf{e}$$

produces an equation with the scaling indicators as explanatory variables and composite error

$$u = \varepsilon - \mathbf{a}^\top\mathbf{e}.$$

The explanatory variables generally correlate with the composite error because both contain the scaling indicator measurement errors.

After centering and stacking observations, the transformed equation is

$$\mathbf{d} = \mathbf{X}\mathbf{a} + \mathbf{u},$$

where $\mathbf{d} \in \mathbb{R}^n$ and $\mathbf{X} \in \mathbb{R}^{n \times k}$ contain the centered dependent observations and scaling indicator observations defined above, and $\mathbf{u} \in \mathbb{R}^n$ contains the centered composite errors.

The hypothesized model identifies observed instruments $\mathbf{z} \in \mathbb{R}^\ell$ satisfying

$$\mathrm{Cov}(\mathbf{z}, u) = \mathbf{0}, \qquad \mathrm{rank}\{\mathrm{Cov}(\mathbf{z}, \mathbf{x})\} = k, \qquad \ell \geq k.$$

The first condition concerns instrument validity and the second instrument relevance.

Let $\mathbf{Z} \in \mathbb{R}^{n \times \ell}$ contain the centered instrument observations. The first stage regresses the explanatory variables on the instruments:

$$\widehat{\mathbf{X}} = \mathbf{P}\mathbf{X}, \qquad \mathbf{P} = \mathbf{Z}(\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top \in \mathbb{R}^{n \times n}.$$

The second stage regresses the dependent variable on these fitted values:

$$\widehat{\mathbf{a}}_{\mathrm{MIIV}} = (\widehat{\mathbf{X}}^\top\widehat{\mathbf{X}})^{-1}\widehat{\mathbf{X}}^\top\mathbf{d} = (\mathbf{X}^\top\mathbf{P}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{P}\mathbf{d}.$$

The matrices to be inverted are assumed nonsingular, and the loadings and slopes are not subject to equality constraints.

Both approaches therefore use fitted predictors in a least squares regression. JS and JSA construct them through an estimated reliability adjustment, whereas MIIV constructs them by projecting the scaling indicators onto model-implied instruments.

Estimation of Variance, Covariance, and Intercept Parameters

After estimating the factor loadings and latent regression slopes, the remaining variance and covariance parameters can be estimated using several least squares procedures (Maydeu-Olivares et al., in press). For all three estimators, we use reweighted least squares, the default in the implementations considered, holding the estimated loadings and slopes fixed throughout the iterations. Let

$$\widehat{\mathbf{H}} = \widehat{\mathbf{\Lambda}}(\mathbf{I}_q - \widehat{\mathbf{B}})^{-1} \in \mathbb{R}^{p \times q},$$

where $\widehat{\mathbf{\Lambda}}$ and $\widehat{\mathbf{B}}$ are obtained using JS, JSA, or MIIV, assuming that $\mathbf{I}_q - \widehat{\mathbf{B}}$ is nonsingular. Conditional on these estimates, the implied covariance matrix is

$$\mathbf{\Sigma}(\mathbf{v}) = \widehat{\mathbf{H}}\mathbf{\Psi}(\mathbf{v})\widehat{\mathbf{H}}^\top + \mathbf{\Theta}(\mathbf{v}),$$

where $\mathbf{v} \in \mathbb{R}^{d_v}$ collects the free variance and covariance parameters.

Conditional on the estimated loadings and slopes, the implied covariance moments are linear in these parameters (Ernst et al., 2023). Incorporating fixed entries and linear equality restrictions gives

$$\mathrm{vech}\{\mathbf{\Sigma}(\mathbf{v})\} = \widehat{\mathbf{c}} + \widehat{\mathbf{G}}\mathbf{v},$$

where $t = p(p+1)/2$, $\widehat{\mathbf{c}} \in \mathbb{R}^t$ is the fixed component under the chosen parameterization, and $\widehat{\mathbf{G}} \in \mathbb{R}^{t \times d_v}$ maps the free parameters to the implied covariance moments.

For a weight matrix $\widehat{\mathbf{W}} \in \mathbb{S}^t_{++}$, the weighted least squares estimate is

$$\widehat{\mathbf{v}} = \left(\widehat{\mathbf{G}}^\top\widehat{\mathbf{W}}\widehat{\mathbf{G}}\right)^{-1}\widehat{\mathbf{G}}^\top\widehat{\mathbf{W}}\left\{\mathrm{vech}(\mathbf{S}) - \widehat{\mathbf{c}}\right\},$$

assuming that $\widehat{\mathbf{G}}$ has full column rank.

Reweighted least squares updates the normal theory weight matrix using the current implied covariance matrix (Browne, 1974; Lee & Jennrich, 1979):

$$\widehat{\mathbf{W}}^{(\tau)} = \frac{1}{2}\mathbf{D}_p^\top \left[\{\widehat{\boldsymbol{\Sigma}}^{(\tau)}\}^{-1} \otimes \{\widehat{\boldsymbol{\Sigma}}^{(\tau)}\}^{-1}\right] \mathbf{D}_p, \qquad \widehat{\boldsymbol{\Sigma}}^{(\tau)} = \boldsymbol{\Sigma}(\widehat{\mathbf{v}}^{(\tau)}).$$

Here, $\otimes$ denotes the Kronecker product, and $\mathbf{D}_p \in \mathbb{R}^{p^2 \times t}$ is the duplication matrix satisfying

$$\operatorname{vec}(\mathbf{S}) = \mathbf{D}_p \operatorname{vech}(\mathbf{S}).$$

At iteration τ, substituting $\widehat{\mathbf{W}}^{(\tau)}$ into the weighted least squares expression gives $\widehat{\mathbf{v}}^{(\tau+1)}$. This procedure is repeated until convergence, assuming that the implied covariance matrices used to construct the weights remain positive definite. The resulting $\widehat{\mathbf{v}}$ determines $\widehat{\boldsymbol{\Theta}}$ and $\widehat{\boldsymbol{\Psi}}$, which are not constrained to be positive semidefinite.

Finally, the intercepts are estimated from the sample means, using the previously estimated factor loadings and latent regression slopes. Under the scaling indicator parameterization, the mean structure is saturated when all nonscaling indicator intercepts and all structural intercepts are free. Let $\overline{\mathbf{m}} \in \mathbb{R}^q$ contain the sample scaling indicator means, ordered by latent variable. Then

$$\widehat{\boldsymbol{\mu}}_\eta = \overline{\mathbf{m}}, \qquad \widehat{\boldsymbol{\nu}} = \overline{\mathbf{y}} - \widehat{\boldsymbol{\Lambda}}\overline{\mathbf{m}} \in \mathbb{R}^p, \qquad \widehat{\boldsymbol{\alpha}} = (\mathbf{I}_q - \widehat{\mathbf{B}})\overline{\mathbf{m}} \in \mathbb{R}^q.$$

These expressions preserve the fixed zero scaling indicator intercepts and reproduce the observed sample means. Additional mean structure restrictions require a corresponding constrained estimation step.

Together, these stages estimate the free entries of $\boldsymbol{\Lambda}$, $\mathbf{B}$, $\boldsymbol{\Theta}$, $\boldsymbol{\Psi}$, $\boldsymbol{\nu}$, and $\boldsymbol{\alpha}$. Although the estimated factor loadings and latent regression slopes are held fixed during covariance estimation, their sampling uncertainty is propagated when calculating standard errors.

Analytical Standard Errors

James–Stein Estimation

Burghgraeve et al. (2021) derived a variance expression for the latent regression slopes that accounts for the estimated reliability adjustment and used bootstrap standard errors in their evaluations. Here, analytical standard errors are obtained by differentiating the complete estimation procedure with respect to the sample moments. For $T \in \{\text{JS}, \text{JSA}\}$, write

$$\widehat{\boldsymbol{\theta}}_T = g_{n,T}(\mathbf{s}),$$

where $g_{n,T}$ maps the h sample moments to the d parameter estimates. This mapping includes the preliminary error variance estimates, the estimates of the factor loadings and latent regression slopes, the subsequent variance and covariance estimates, and the intercept estimates.

Suppose

$$\sqrt{n}\{\mathbf{s} - \boldsymbol{\sigma}(\boldsymbol{\theta})\} \xrightarrow{d} N_h(\mathbf{0}, \boldsymbol{\Gamma}), \qquad \boldsymbol{\Gamma} \in \mathbb{S}_+^h,$$

where $\boldsymbol{\Gamma}$ is the joint asymptotic covariance matrix of the sample means and covariances. Under the required moment and differentiability conditions, the delta method covariance estimator is

$$\widehat{\mathbf{V}}_T = \frac{1}{n}\widehat{\mathbf{J}}_T \widehat{\boldsymbol{\Gamma}} \widehat{\mathbf{J}}_T^\top \in \mathbb{S}_+^d, \qquad \widehat{\mathbf{J}}_T = \left.\frac{\partial g_{n,T}(\mathbf{s})}{\partial \mathbf{s}^\top}\right|_{\mathbf{s}=\mathbf{s}_{\text{obs}}} \in \mathbb{R}^{d \times h}.$$

The Jacobian propagates uncertainty through all estimation stages, including the intercept estimates. It is evaluated analytically by default, including implicit differentiation of the converged reweighted least squares equations. For JSA, it also incorporates derivatives of the estimated aggregation weights and normalization loadings.

The matrix $\boldsymbol{\Gamma}$ can be estimated using normal theory or empirical moments through order four (Browne, 1984). Normal theory estimates may be evaluated at the sample covariance matrix or the fitted model covariance matrix. For parameter $r = 1, \ldots, d$, the analytical standard error is

$$\widehat{\mathrm{SE}}(\widehat{\theta}_{T,r}) = \sqrt{[\widehat{\mathbf{V}}_T]_{rr}} \in [0, \infty).$$

Model-Implied Instrumental-Variable Estimation

For the MIIV estimates of the factor loadings and latent regression slopes in one equation, the conventional homoskedastic covariance estimator is

$$\widehat{\mathbf{V}}_{\mathbf{a},\mathrm{MIIV}} = \widehat{\sigma}_u^2 (\mathbf{X}^\top \mathbf{P} \mathbf{X})^{-1} \in \mathbb{S}_+^k,$$

where

$$\widehat{\sigma}_u^2 = \frac{\widehat{\mathbf{u}}^\top \widehat{\mathbf{u}}}{n - k - 1}, \qquad \widehat{\mathbf{u}} = \mathbf{d} - \mathbf{X}\widehat{\mathbf{a}}_{\mathrm{MIIV}}.$$

The intercept is absorbed by centering, and $n > k+1$ is assumed. Here, σ_u^2 is the variance of the composite error in the transformed equation, rather than an individual measurement error or structural disturbance variance. This is the default covariance estimator for the MIIV loadings and slopes in the implementation considered.

For the variance and covariance parameters $\mathbf{v}$, the delta method accounts for uncertainty in the previously estimated loadings and slopes (Bollen & Maydeu-Olivares, 2007; Jin et al., 2021). Define

$$\widehat{\mathbf{H}}_2 = (\widehat{\mathbf{G}}^\top \widehat{\mathbf{W}} \widehat{\mathbf{G}})^{-1} \widehat{\mathbf{G}}^\top \widehat{\mathbf{W}} \in \mathbb{R}^{d_v \times t},$$

evaluated at the converged weights and treated as fixed in the linearization. Let d_a denote the total number of free factor loadings and latent regression slopes. Let $\widehat{\boldsymbol{\Delta}}_1 \in \mathbb{R}^{t \times d_a}$ be the Jacobian of the implied covariance moments with respect to these parameters, evaluated at the estimates, and let $\widehat{\mathbf{K}} \in \mathbb{R}^{d_a \times t}$ be the Jacobian of their MIIV estimates with respect to $\mathrm{vech}(\mathbf{S})$ (Fisher & Bollen, 2020).

Under correct specification, the covariance estimator is

$$\widehat{\mathbf{V}}_{\mathbf{v},\mathrm{MIIV}} = \frac{1}{n} \widehat{\mathbf{H}}_2 (\mathbf{I}_t - \widehat{\boldsymbol{\Delta}}_1 \widehat{\mathbf{K}}) \widehat{\boldsymbol{\Gamma}}_S (\mathbf{I}_t - \widehat{\boldsymbol{\Delta}}_1 \widehat{\mathbf{K}})^\top \widehat{\mathbf{H}}_2^\top \in \mathbb{S}_+^{d_v},$$

where $\widehat{\boldsymbol{\Gamma}}_S \in \mathbb{S}_+^t$ estimates the asymptotic covariance matrix of the sample covariance moments. It is the covariance-moment block of $\widehat{\boldsymbol{\Gamma}}$. The factor

$$\mathbf{I}_t - \widehat{\boldsymbol{\Delta}}_1 \widehat{\mathbf{K}}$$

propagates the sampling uncertainty of the estimated factor loadings and latent regression slopes through the residual covariance moments.

Moment based covariance estimates are also available for the MIIV factor loadings and latent regression slopes as

$$\frac{1}{n} \widehat{\mathbf{K}} \widehat{\boldsymbol{\Gamma}}_S \widehat{\mathbf{K}}^\top.$$

For the intercepts, moment based standard errors follow by differentiating the mean structure expressions with respect to the sample moments. These calculations provide delta method standard errors separately for each parameter block. In particular, the second stage expression uses the converged weights as fixed in its linearization, whereas the JS and JSA Jacobians differentiate the complete estimation procedure. A related delta method calculation for instrumental variable regression is presented by Maydeu-Olivares et al. (2019).

Simulation Design and Performance Evaluation

In this section we describe the design of the simulation study and the measures used to evaluate the standard errors.

Population Model and Estimands

We study the model in Figure 1, with three latent variables and four indicators each. The latent variable model is

$$\begin{aligned} \eta_2 &= \alpha_2 + B_{21}\eta_1 + \zeta_2, \\ \eta_3 &= \alpha_3 + B_{31}\eta_1 + B_{32}\eta_2 + \zeta_3. \end{aligned} \tag{1}$$

The design is partly based on previous studies. Burghgraeve et al. (2021) used four indicators per latent variable, a common reliability of .6 or .8 for all indicators. Note also that the model is similar to the empirical illustration in Burghgraeve et al. (2021). Moreover, Jin et al. (2021) used a recursive latent variable model with several equations and standardized slopes between 0.1 and 0.6, which the present model follows partly with $B_{21} = 0.6$, $B_{31} = 0.3$, and $B_{32} = 0.5$. With $\mathrm{Var}(\eta_1) = 1$, $\mathrm{Var}(\zeta_2) = 0.64$, and $\mathrm{Var}(\zeta_3) = 0.48$, the variances of η_2 and η_3 equal 1, so that the slopes are standardized, and the implied correlations are .60 between η_1 and η_2, .60 between η_1 and η_3, and .68 between η_2 and η_3. The scaling indicators have loadings fixed to one and the remaining loadings range from 0.8 to 1.2. The mean structure has $E(\eta_1) = 1$, $\alpha_2 = \alpha_3 = 0.5$, and intercepts of 0.2 for the nonscaling indicators. The main estimands are all free factor loadings, latent regression slopes, residual variances, and latent variances and disturbance variances.

Simulation Conditions and Estimation Approaches

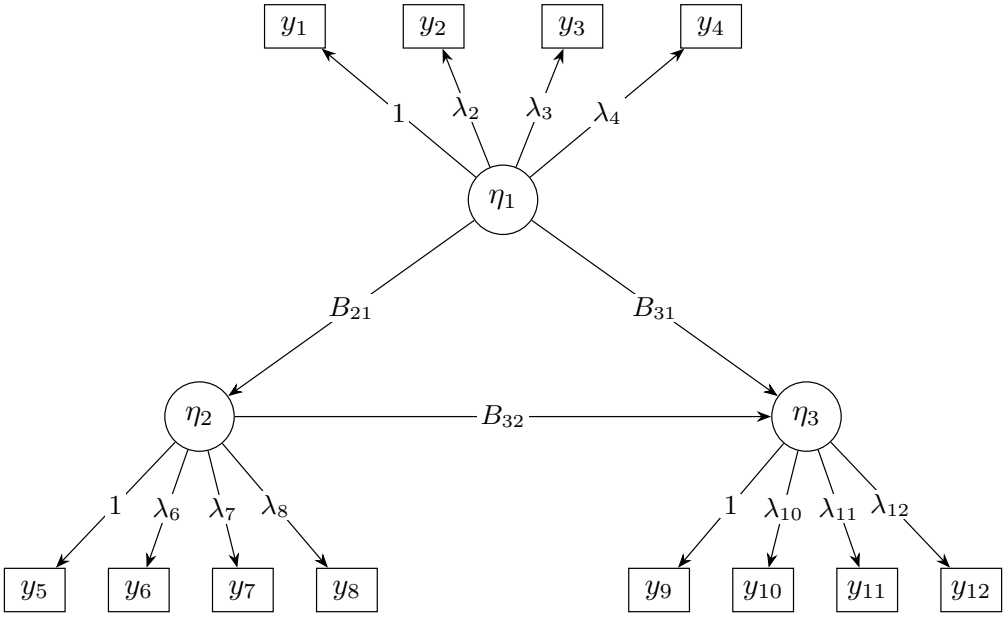
We employed a fully factorial design with two factors: sample size ($n \in \{100, 200, 300, 400, 500, 600\}$) and indicator reliability ($\rho \in \{.6, .8\}$). This yielded $6 \times 2 = 12$ conditions, each replicated 1,500 times. The sample sizes cover the small to moderate range in which the equation by equation estimators are expected to differ most from ML. The reliability levels match the low and high conditions of Burghgraeve et al. (2021) and determine the size of the reliability adjustment in JS and JSA. As highlighted throughout the article, we investigated the JS, JSA, MIIV, and ML approaches, all with the default options in lavaan (Rosseel, 2012).

Data Generation

Data were generated from the population model with all stochastic components normal and mutually independent. The exogenous latent variable was generated as $\eta_1 \sim \mathcal{N}(1, 1)$ and the structural disturbances as $\zeta_2 \sim \mathcal{N}(0, 0.64)$ and $\zeta_3 \sim \mathcal{N}(0, 0.48)$, from which η_2 and η_3 follow by Equation 1. The measurement error of each indicator was generated as normal with mean zero and variance $\lambda^2(1 - \rho)/\rho$, where λ is the loading of that indicator, so that every indicator has reliability ρ in its condition, and the indicators follow by the measurement equations. The observed variables are therefore multivariate normal with the mean vector and covariance matrix implied by the population values.

Figure 1

Path diagram of the simulation model with three latent variables and their indicators.



Note. Scaling indicator loadings are fixed to one. The population latent regression slopes are $B_{21} = 0.6$, $B_{31} = 0.3$, and $B_{32} = 0.5$. Intercepts, measurement errors, disturbances, and variances are omitted for clarity.

Convergence, Outliers, and Inadmissible Solutions

Replications were screened in two steps, and a replication that failed either step for any estimator was removed for all four. A solution was nonconverged when any estimate, standard error, or confidence interval bound of an estimand was missing or non-finite. Among the converged replications, outliers were point estimates more than three interquartile ranges below the first or above the third quartile, per condition, estimator, and estimand. Inadmissible solutions, with a nonpositive variance estimate, were retained and counted.

Performance Measures

Performance was assessed with the measures implemented in the simhelpers package (Joshi & Pustejovsky, 2025), computed for every estimand. These measures, with their empirical and Monte Carlo standard error (MCSE) calculations, are listed in Table 1. The MCSEs of the SD and of the SE^2/VAR ratio are obtained by jackknife and that of coverage in closed form. The SE/SD ratio and the SE^2/VAR ratio compare the same quantities on the standard deviation and on the variance scale, so only the latter, for which an MCSE is available, is shown in the figures. Coverage between .925 and .975 and SE^2/VAR ratios between 0.90 and 1.10 were considered acceptable.

Although the evaluation targets the standard errors, the relative bias of the estimates was also computed, because standard errors of markedly biased estimates are of limited use for inference. It is reported in the text, but not in the figures.

Reproducibility and Computational Details

All analyses were conducted in R (R Core Team, 2026) using the lavaan package (Rosseel, 2012) for estimation, simhelpers (Joshi & Pustejovsky, 2025) for performance evaluation, and ggplot2 (Wickham, 2016) for the figures. The computational environment was pinned with Nix (Dolstra et al., 2004) via the rix package (Rodrigues & Baumann, 2026), and a Dockerfile that builds the same environment inside a container

Table 1

is also provided. The article was written with the apaquarto extension (Schneider, 2026). For an extensive description of all files and how to reproduce the simulation study as well as the manuscript, see the repository.

Results

Convergence, Outliers, and Inadmissible Solutions

All four estimators converged in every replication of every condition. Outlier screening removed 47 of the 18,000 replications (0.26%), concentrated at $n = 100$ with reliability .6, where 1.9% of the replications were removed, flagged mainly through JS (1.7% of the replications in that condition). In every other condition fewer than 0.5% of the replications were removed. Inadmissible solutions occurred in 4 replications, all for JS at $n = 100$ with reliability .6, where a disturbance variance estimate was negative. All of them were also flagged by the outlier screening, so none enters the results below.

Relative Bias

JS, JSA, and ML were essentially unbiased, with absolute relative bias of at most 4.5% for any parameter and condition. MIIV was also nearly unbiased except at reliability .6 and small samples. At $n = 100$ its loadings were slightly underestimated, by 6.2 to 6.9%, whereas its latent and disturbance variances were overestimated by 11.0 to 12.4%. From $n = 300$ onward, all estimators were within 4.6% of the population values.

Standard Error Performance

Each figure presents reliability .6 in the upper half and .8 in the lower half. Figure 2 shows the latent regression slopes, Figure 3 one loading per latent variable, Figure 4 the residual variances of the scaling indicators and one nonscaling indicator per latent variable, and Figure 5 the variance of η_1 and the two disturbance variances. Omitted loadings and residual variances behave similarly to those shown for the same latent variable, with SE^2/VAR ratios that differ by at most 0.15 and coverage by .03 in every condition.

Overall, standard error performance was acceptable. Across all conditions, estimators, and parameters, 96.3% of coverage rates fell between .925 and .975 and 97.9% of SE^2/VAR ratios between 0.90 and 1.10, with no ratio below 0.90. From $n = 300$ onward, the few ratios above 1.10 remained below 1.12. Three departures emerge at $n = 100$ and $n = 200$, mainly at reliability .6.

First, JS standard errors for B_{31} and B_{32} exceed the acceptable range at reliability .6, with SE^2/VAR ratios of 1.5–1.6 at $n = 100$ and 1.13–1.14 at $n = 200$. Coverage nonetheless exceeds .975 only for B_{31} at $n = 100$. Its SD is also the largest, 0.24 versus 0.13–0.17 at $n = 100$. These patterns may reflect instability in the reliability adjustment for the η_3 equation. In some replications, the implied correlation between η_1 and η_2 approaches 1, making the fitted predictors nearly collinear and inflating both estimates and standard errors. Screening removed the most extreme cases, but milder cases remain. JSA, which aggregates indicators within each latent variable, shows no corresponding instability.

Second, MIIV loading intervals fall below the acceptable coverage range at reliability .6, with coverage of .88–.90 at $n = 100$, compared with .93–.96 for the other estimators, despite SE^2/VAR ratios of 0.98–1.14 and the smallest SD. This discrepancy reflects the downward loading bias reported above, consistent with the weak instrument effect described by Jin et al. (2021), and diminishes slowly as sample size increases.

Third, at $n = 100$ MIIV overestimates the standard errors of the latent and disturbance variances, with ratios of 1.11–1.18, but its coverage rates of .96–.97 remain within the acceptable range. ML, in contrast, has ratios close to 1 but coverage rates of .91–.94 for the disturbance variances. JS and JSA fall between them. At reliability .8, the coverage rate for the disturbance variance of η_2 is about .92 for ML, JS, and JSA, just

below the acceptable range. Residual variances show less contrast, with coverage rates from .91 to .96 across estimators and a common dip at $n = 100$.

Discussion

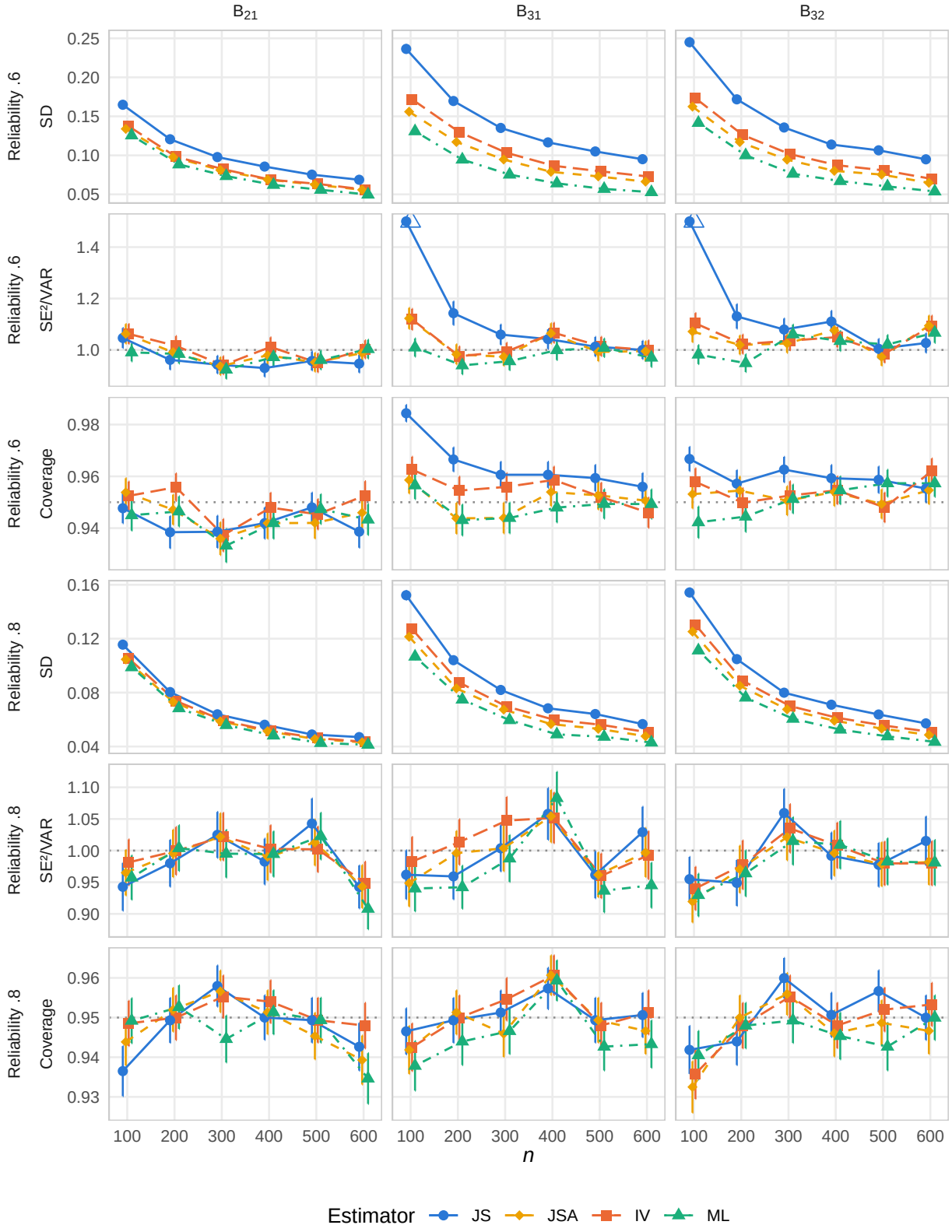
This article described MIIV, JS, and JSA estimation as currently implemented in the R package lavaan. It also developed and illustrated analytical standard errors for JS and JSA. A reproducible simulation study evaluated their standard error accuracy against ML under correct model specification.

The findings broadly align with the studies that motivated the simulation design. Bollen (2019), summarizing Bollen et al. (2007), reports few differences between MIIV and ML under correct specification with normally distributed data. Our results agree in terms of bias and standard error accuracy from $n = 300$ onward, although MIIV remained less precise than ML for the latent regression slopes. Jin et al. (2021) found that low scaling indicator reliability can lead to biased MIIV estimates in small samples and that standard errors for variance parameters were the least accurate. Related patterns appeared here at reliability .6 and $n = 100$, with loading bias and undercoverage, alongside inflated standard errors for latent and disturbance variances. Burghgraeve et al. (2021) found JS less efficient than JSA for latent regression estimation and JSA competitive with ML, consistent with the empirical standard deviations observed here. However, their bootstrap standard errors for JS were too small in small samples, whereas our analytical standard errors were too large at reliability .6 and $n \leq 200$. Whether this difference reflects the standard error methods or differences in the simulation conditions requires further investigation.

The simulation was intended as a focused evaluation rather than an exhaustive comparison. Further research should compare MIIV, JS, and JSA across a broader range of conditions, including departures from normality and misspecification of the measurement and latent variable models.

Figure 2

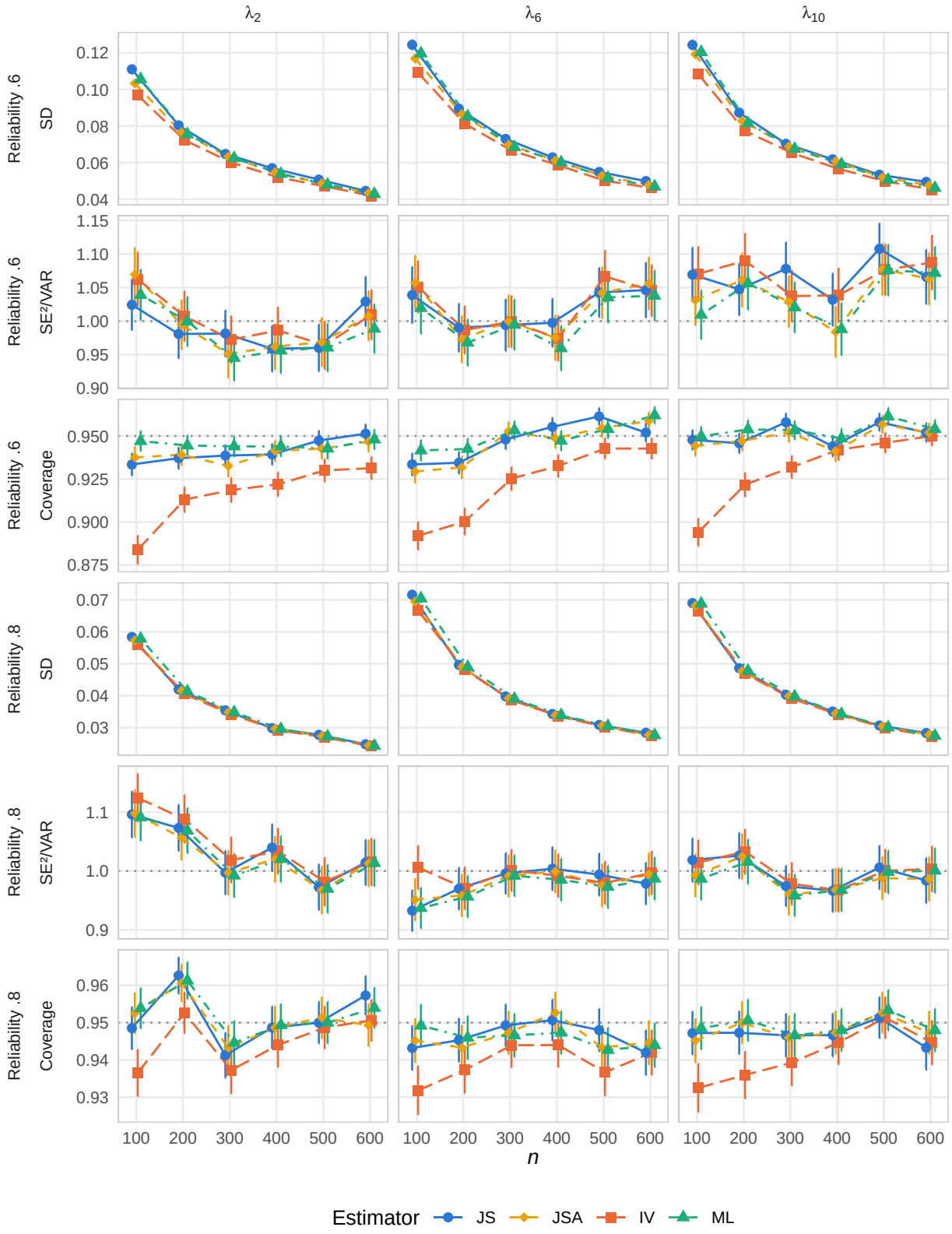
Performance of the Standard Errors of the Latent Regression Slopes



Note. JS = scaling James–Stein, JSA = aggregated James–Stein, IV = MIIV, ML = maximum likelihood. Dotted lines mark the reference values 1 and .95. Error bars represent Monte Carlo standard errors. Values above 1.5 are drawn at 1.5 with an open triangle.

Figure 3

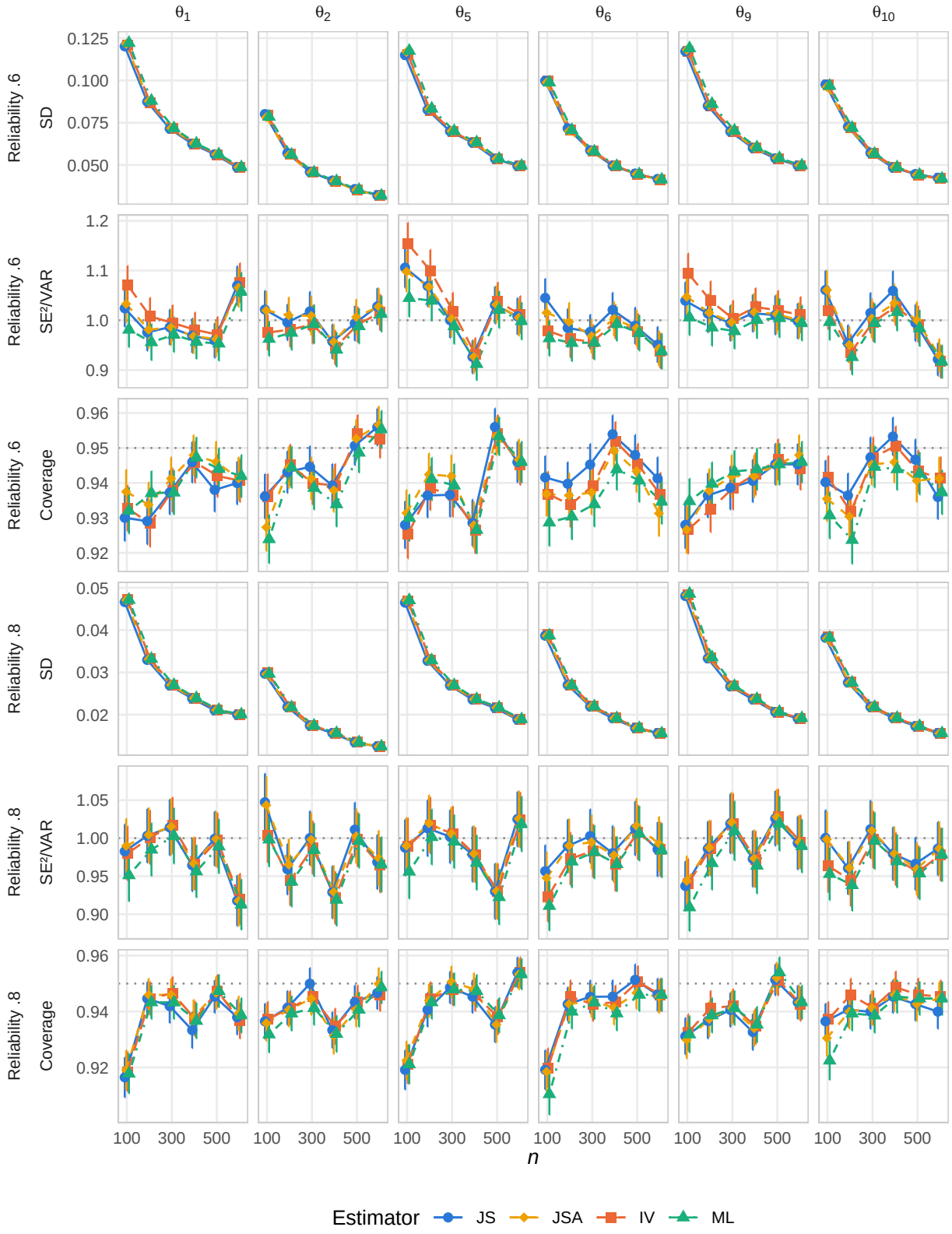
Performance of the Standard Errors of One Factor Loading per Latent Variable



Note. JS = scaling James–Stein, JSA = aggregated James–Stein, IV = MIIV, ML = maximum likelihood. Dotted lines mark the reference values 1 and .95. Error bars represent Monte Carlo standard errors. Values above 1.5 are drawn at 1.5 with an open triangle.

Figure 4

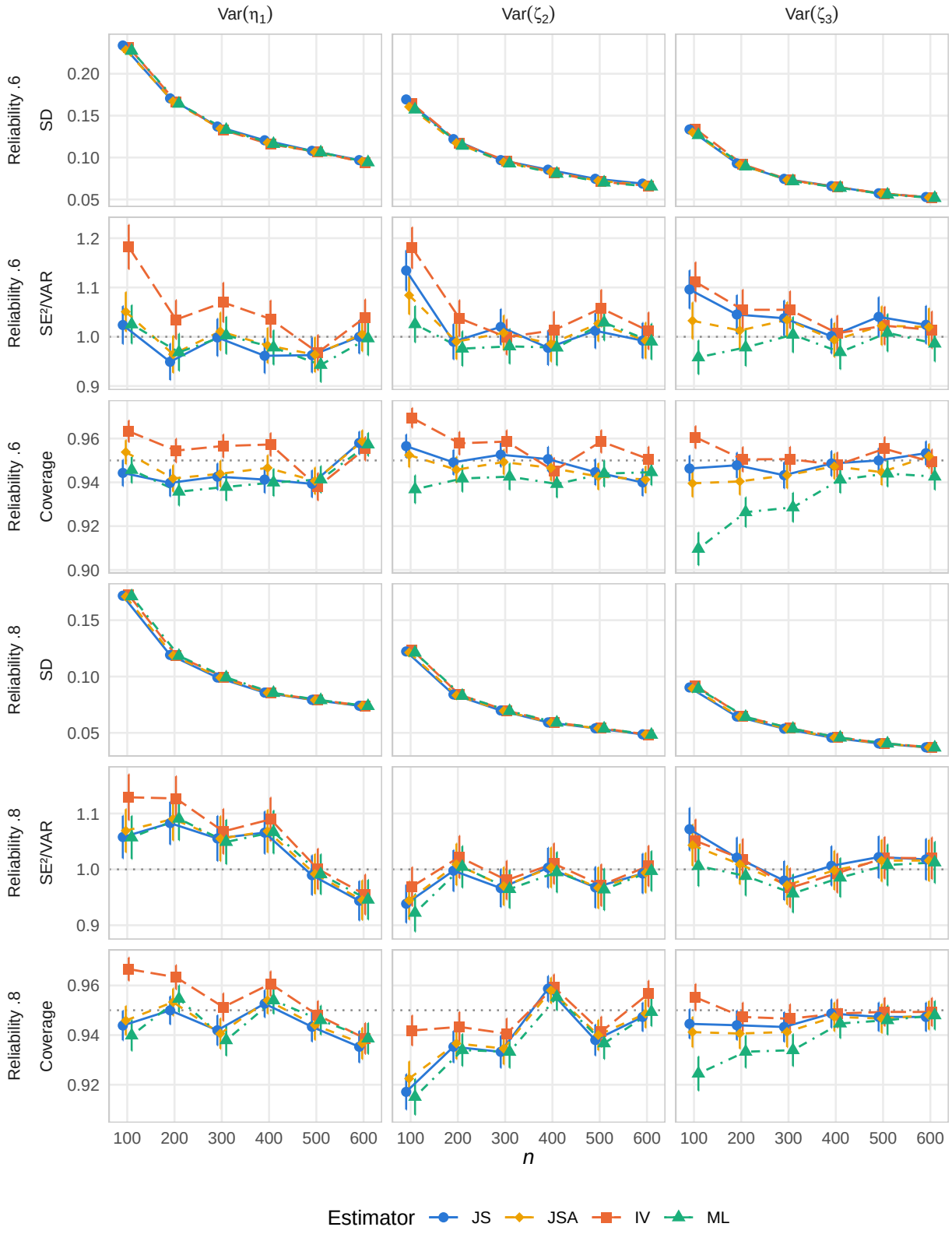
Performance of the Standard Errors of the Residual Variances of the Scaling Indicator and One Nonscaling Indicator per Latent Variable



Note. JS = scaling James–Stein, JSA = aggregated James–Stein, IV = MIIV, ML = maximum likelihood. Dotted lines mark the reference values 1 and .95. Error bars represent Monte Carlo standard errors. Values above 1.5 are drawn at 1.5 with an open triangle.

Figure 5

Performance of the Standard Errors of the Latent and Disturbance Variances



Note. JS = scaling James–Stein, JSA = aggregated James–Stein, IV = MIIV, ML = maximum likelihood. Dotted lines mark the reference values 1 and .95. Error bars represent Monte Carlo standard errors. Values above 1.5 are drawn at 1.5 with an open triangle.

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